



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

Document Type Standard Operating Procedure	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 1 of 3
Document Code: SOP-09-01	QUALITY CONTROL CHART GUIDE		

Introduction

This document provides the guidelines on how to come up with a Quality Control Chart.

1. Responsibilities

1. Technical Manager
2. Quality Assurance Manager
3. Laboratory Analyst

2. GATHERING DATA FOR A QUALITY CONTROL CHART

2.1 Tabulate data for a control chart following the format set in LF 09-01, Quality Control Table.

2.2 The table must have a minimum of 20 points. For analytical methods, these points may come from validation data. Calculate the following using the formulae set in Table 1 below:

- mean value, which shall be the central line of the control chart
- standard deviation of the mean.
- warning limits
- control limits

3. CONSTRUCTING THE QUALITY CONTROL CHART

3.1 Construct a graph with the run number as the x axis and the value of the controlled parameter (for histamine, the concentration/percent recovery) as the y.

3.2 Set the mean as the central line of the plot.

3.3 Set the warning and control limits as straight lines in the plot.

3.4 Plot the quality control data obtained according to SOP-09 (Ensuring the Validity of Results) in the graph.

Prepared by: MA. MAURICE A. BANQUILING Quality Assurance Manager	Reviewed and Approved by: EUGENE GAY B. JAMORA Laboratory Manager
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Table 1. Important mathematical formulae

Mean (μ)	$\frac{\text{Sum of results}}{\text{Number of analysis}}$
Standard Deviation (s)	$\sqrt{((\Sigma(x - \mu)^2)/n - 1)}$
Upper Warning Limit (UWL)	$\mu + 2s$
Lower Warning Limit (LWL)	$\mu - 2s$
Upper Control Limit (UCL)	$\mu + 3s$
Lower Control Limit (LCL)	$\mu - 3s$

4. CRITICAL POINTS

4.1 Attention must be paid and preventive actions set in motion, if applicable, when

- a data point falls outside the warning limits
- if five successive points are on the same side of the central line
- if five successive points create an ascending or descending line.

4.2 Corrective actions must be initiated when

- a data point falls outside the control limits
- five successive points fall outside the warning limits
- if seven successive points are on the same side of the central line
- if seven successive points create an ascending or descending line
- 14 consecutive points are alternating up and down

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Document Code: SOP-09-01	QUALITY CONTROL CHART GUIDE		

- 2 out of 3 consecutive points are more than 2 sigmas from the center line in the same direction
- 4 out of 5 consecutive points are more than 1 sigma from the center line in the same direction
- 15 consecutive points are within 1 sigma of the center line
- 8 consecutive points on either side of the center line with not within 1 sigma
- Any noticeable or predictable pattern, cycle, or trend

5. Forms

LF 10-01, Quality Control Table

6. References

EURACHEM Guide Fitness for Purpose

7. Document History

Revision No.	Issue Date	Original Entry	Revised As	Recall Date

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